Wei Wei

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RESEARCH SUMMARY

Ph.D. student in Computer Science researching **data visualization and human-computer interaction**, with a focus on **perception + visualization**, **visualization co-design**, and **data physicalization**. Research experience since 2019, including research internships in Canada and France and first-author publications in **IEEE TVCG** and **IEEE VIS**.

EDUCATION

University of Victoria — **Victoria, Canada**
Ph.D. in Computer Science — *May 2023 - Present (Expected Sep 2027)*
First-year CGPA: 8.8/9.0

University of Calgary — **Calgary, Canada**
M.Sc. in Computer Science — *Jan 2021 - May 2023*
CGPA: 4.00/4.00 Thesis: *Design of Anthropomorphic Interfaces for Autonomous Vehicle-Pedestrian Interaction*

Southwest Jiaotong University — **Chengdu, China**
B.Sc. in Computer Science and Technology — *Sept 2016 - Jun 2020*
CGPA: 3.51/4.00

PUBLICATIONS

- **Wei, W.**, Nacenta, M., Miranda, M., & Perin, C. (2025). *Locatability and Locatability Robustness of Visual Variables in Single Target Localization*. **IEEE Transactions on Visualization and Computer Graphics (TVCG)**. [19% acceptance rate]
- **Wei, W.**, Huron, S., & Jansen, Y. (2023). *Towards Autocomplete Strategies for Visualization Construction*. **IEEE VIS 2023** Short Paper.
- Danyluk, K., Ulusoy, T., **Wei, W.**, & Willett, W. (2020). *Touch and Beyond: Comparing Physical and Virtual Reality Visualizations*. **IEEE Transactions on Visualization and Computer Graphics (TVCG)**.
- **Wei, W.**, Daneshzand, F., Wang, Z., Mattson, E., Farkas, J., Storteboom, S., Perin, C., & Carpendale, S. (2026). *Epistemic Tensions: Reframing A Visualization Co-Design through Entanglement Theory*. IEEE VIS 2026 BELIV Workshop paper.
- Huron, S., **Wei, W.**, Tseng,W., Bonnail,E., & Genest, E. (2026) *This is not a speculation, this is the reality: Reflection on Speculative Design Workshop in HCI & Design Research*. Workshop paper at the ACM Conference on Human Factors in Computing Systems (CHI).
- **Wei, W.**, Daneshzand, F., Wang, Z., Mattson, E., Perin, C., & Carpendale, S. (2025). *The Fuzzy Front Ends: Reflections on The Never-Ending Story of Visualization Co-Design*. IEEE VIS 2025 alt.VIS Workshop paper.
- **Wei, W.**, Daneshzand, F., Wang, Z., Mattson, E., Perin, C., & Carpendale, S. (2026). *The Dark Sides of Data Visualization: Critical Reflections from a Co-Design Practice*. Poster presented at the IEEE Pacific Visualization Conference, Sydney, Australia.
- **Wei, W.**, Perin, C., & Nacenta, M. (2026). *Investigating Time Costs of Target Localization and Difference Estimation in Visualization*. Poster presented at the IEEE Pacific Visualization Conference, Sydney, Australia.

RESEARCH EXPERIENCE

Research Intern, AVIZ, Université Paris-Saclay, Paris, France *May 2026 - Present*
- Developed a user study on reVISit2 (JavaScript and React) to evaluate how motion in sports footage affects the perception of embedded visualizations.
- Compared embedded bar and donut charts across basketball, tennis, and swimming videos with crowd-sourced participants; analyzed the data using Bayesian statistics.

Research Assistant, VIXI Lab, University of Victoria *May 2023 - Present*
- Developed and deployed a web-based visualization perception experiment with 224 participants.
- Analyzed the results using frequentist and Bayesian statistics in R and Python; published first-author findings on the locatability of visual variables in IEEE TVCG.
- Built a multimodal analysis workflow in Python and JupyterLab that uses LLMs to automate the analysis of eye movement patterns in eye-tracking data.

Research Assistant, University of Victoria & Simor Fraser University *Jan 2024 - Jan 2026*
- Developed an interactive web-based visualization system using D3.js to analyze Canadian arts funding data.
- Surfaced epistemic tensions in community-centered visualization co-design practice, contributing design knowledge and methodological reflections in first-author papers at BELIV and IEEE VIS alt.VIS.

Research Intern, Institut Polytechnique de Paris *Aug 2022 - Dec 2022*
- Conducted a speculative workshop with 15 participants to investigate autocomplete strategies for visualization construction.
- Analyzed participants' sketches and transcripts using thematic analysis, leading to a first-author IEEE VIS short paper.

Graduate Research Assistant, iLab, University of Calgary *Jan 2021 - May 2023*
- Member of the uTouch Group, Interactions Lab; completed an M.Sc. thesis on anthropomorphic interfaces for autonomous vehicle-pedestrian interaction.

Research Intern, University of Calgary *Jul 2019 - Oct 2019*
- Advised by Wesley Willett and Tony Tang; contributed to research comparing physical and virtual reality visualizations, published in IEEE TVCG.

SKILLS

Programming	JavaScript, Python, R, C++, C, C#, Java, Lua
Frameworks & Tools	D3, reVISit 2, Jupyter Notebook, Unity3D, Oculus VR, LaTeX
Research Methods	User study design and deployment, co-design, Bayesian data analysis, thematic analysis

SELECTED AWARDS

- Mitacs Globalink Research Award, 2026
- University of Victoria Fellowship, 2023–2025
- Mitacs Globalink Research Award, 2022
- Departmental Research Award, University of Calgary, 2021
- Mitacs Globalink Graduate Fellowship, 2021
- China Scholarship Council funding for research abroad, 2019

TEACHING AND SERVICE

- Graduate Teaching Assistant, CPSC 231 (Introduction to Computer Science I), University of Calgary, May 2021 - May 2022 (2 terms)
- Organizer, UBCV-UVIC Hackathon, Jan 2024 - May 2024
- Student volunteer chair of IEEE VIS 2026 Paris Satellite event, Nov 2026